Disclosure Regulation and Price Informativeness: Evidence from Industry-Information Disclosure Guidelines in China

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ABSTRACT We examine whether China's industry information disclosure guidelines (IIDGs) improve firms' stock price informativeness and the quality of their textual disclosure. Employing a stacked difference-in-differences design, we find that applicable firms respond to IIDG implementation by including additional IIDG-related and firm-specific information in the Management Discussion and Analysis (MD&A). Our results also show that the stock market reacts to the enhanced disclosure and incorporates more firm-specific (rather than market-wide or industry-level) information into the stock price, as evidenced by a decrease in stock return synchronicity. Further analyses show that firms with fewer increases in textual disclosure following IIDG implementation are more likely to receive disclosure-related comment letters. Additionally, the extent to which firms change their textual disclosure varies with the industry concentration and firms' profit margins due to the proprietary costs of disclosure.

Keywords: disclosure regulation; textual disclosure; market reaction; stock return synchronicity

JEL codes: G14; G18; G38; M4

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1. Introduction

Information disclosure regulations are critical in financial accounting (Healy & Palepu, 2001; Leuz & Wysocki, 2016). Recent studies in this field predominantly focus on non-financial or narrative information regulation, such as corporate social responsibility (CSR) or key audit matters (e.g., Chen et al., 2018; Sirois et al., 2018; Burke et al., 2023; Gibbons, 2023). Our study seeks to contribute to this line of literature by delving into the effects of industry-tailored disclosure regulation on Chinese public firms' textual disclosure and, consequently, their price informativeness.

In 2013, the Shanghai and Shenzhen Stock Exchanges in China introduced sets of industry information disclosure guidelines (IIDGs). Each set of guidelines addresses a specific industry and requires firms in that industry to disclose useful and firm-specific information.[1] The information includes, but is not limited to, the industry's overall development and the firm's potential risks, future development prospects, business model, detailed operation information, and strategic plan and progress. For example, the IIDG for the e-commerce industry requires e-commerce firms to disclose operational information that is specific to the industry, such as the firm's business model, platform, website, page views, and registered users. In contrast, the IIDG for the real estate industry requires firms to disclose operational information such as the land area of projects under construction, newly started projects, and completed projects.[2] Overall, the IIDGs aim to facilitate Chinese stock market investors' informed decision-making.

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[1] See http://www.sse.com.cn/disclosure/credibility/dynamic/c/c_20160302_4052889.shtml (in Chinese), http://www.sse.com.cn/disclosure/credibility/supervision/dynamic/c/c_20160524_4118135.shtml (in Chinese), http://www.sse.com.cn/aboutus/mediacenter/hotandd/c/c_20161202_4209904.shtml (in Chinese), http://www.szse.cn/aboutus/trends/news/t20150226_518573.html (in Chinese), and http://www.szse.cn/aboutus/trends/news/t20191027_571575.html (in Chinese).

[2] Online Appendix Table I presents a list of all IIDGs and their effective implementation dates. Online Appendix Table II lists the keywords for our four sample IIDGs and the corresponding firms' sample MD&A disclosure changes.

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By defining the specific information that firms in that industry should disclose, the guidelines require firms to unveil a broader range of information than they traditionally would disclose. The disclosure regulation can induce firms to provide more textual disclosure of IIDG-related and firm-specific information in accordance with the relevant IIDG. This information may reduce investors' processing costs and limit managers' disclosure discretion. Market participants are likely to place greater weight on IIDG-required disclosures than they do on other market-wide disclosures. Consequently, investors can better integrate the information and more precisely value the disclosing firms.

We argue that increased weight on IIDG-required disclosure in trading decisions can improve price informativeness to the extent that stock prices reflect the disclosed information (Brunnermeier, 2005; Blankespoor et al., 2020). However, firms are exempt from the disclosure requirements if they have a legitimate reason for being so, although they need to explain that they are exempt.[3] Thus, whether IIDG implementation induces changes to firms' textual disclosure and price informativeness is an open question that should be examined empirically.

Using a sample of Chinese firms listed on the Shanghai and Shenzhen Stock Exchanges from 2009 to 2019, we first provide preliminary support that firms respond to IIDG implementation, as evidenced in the year-over-year changes to MD&As and the MD&As' textual similarity to the relevant IIDG around the implementation event. We then conduct a stacked difference-in-differences (DiD) test and find that IIDG implementation prompts firms to change their textual disclosure by providing more IIDG-

[3] For example, firms with high proprietary costs may be cautious in their disclosure practices and more reluctant to fully comply with the IIDGs.

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related and firm-specific information in their MD&As, proxied by the number of IIDG-related words (certain words and phrases in the IIDG and repeated in the MD&As) and the number of characters in specific entity names in the MD&A scaled by the total character number in the MD&A (Hope et al., 2016). More importantly, we find that firms' price informativeness increases after IIDG implementation, as captured by an increase in the market reaction to the release of firm information and a decrease in the stock return synchronicity. This indicates that the disclosure regulation facilitates the incorporation of more firm-specific information into firms' stock returns. To offer further evidence on IIDGs' impact, we employ the two-stage least squares (2SLS) method and conduct subsample analyses. Our main results are also robust to a battery of robustness tests, including the parallel trend test, the placebo test, the staggered DiD design, and the propensity score matching (PSM) method.

A natural question that arises from our results is why firms are willing to comply with the IIDGs, given that they may have legitimate reasons or exemptions from disclosing according to the guidelines.[4] We answer this question by studying the consequence of not complying with the IIDGs. We demonstrate that firms that increase their textual disclosure less in response to the relevant IIDG are more likely to receive disclosure-related comment letters. This suggests that China's regulatory authorities pay closer attention to firms that fail to adequately comply with the IIDGs. Cross-sectional tests show that firms that operate in more concentrated industries or those that enjoy higher profit margins are less inclined to increase their textual disclosure in response to the guidelines. This finding is consistent with the argument that the threat of potential entrants and proprietary information decrease firms' incentives to provide extensive

[4] We thank the anonymous reviewers for pointing this out.

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disclosure. However, we do not find evidence to support the conjecture that the effect of IIDG implementation on price informativeness varies with either the threat of potential entrants or proprietary costs.

This study contributes to the literature in two ways. First, we contribute to research on disclosure regulations. Prior studies largely focus on disclosure regulations that are related to accounting standards or financial information (Holthausen & Watts, 2001; Kim et al., 2012; Yip & Young, 2012; Hribar et al., 2022). However, a growing body of research pays additional attention to disclosure regulations that involve non-financial or narrative information, such as CSR and key audit matters (e.g., Chen et al., 2018; Sirois et al., 2018; Burke et al., 2023; Gibbons, 2023). We add to this emerging stream of research by investigating the effects of disclosure regulation that specifically targets industrial and operational information. We conduct our research in an emerging market where the need for valuable information is particularly high. Our findings provide compelling evidence that such disclosure regulation offers benefits in enhancing the information content of firm disclosure, as evidenced by the increased textual disclosure in response to IIDGs and the resulting increase in price informativeness.

Second, our study improves our understanding of MD&A disclosure and its role in providing informative financial reporting. As MD&As are important parts of corporate narratives, previous studies explore various MD&A disclosure characteristics such as readability, comparability, and tone, as well as specific sections like risk factors and forward-looking information. These characteristics or specific sections are linked to regulatory requirements or incentives aimed at promoting transparent disclosure and are found to have economic implications for multiple stakeholders, including auditors, peer firms, and debt holders (Lang & Stice-Lawrence, 2015; Muslu et al., 2015; De Franco et al., 2020; Wang, 2020; Cho & Muslu, 2021). Brown and Tucker (2011) and D'Augusta

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and DeAngelis (2020) suggest that MD&As suffer from boilerplate, manipulated, and strategic disclosures. We extend this literature by demonstrating that stock prices are more likely to incorporate the information available in the disclosure when firms are required to disclose information according to the IIDG. This finding suggests that the regulation about the disclosure of industrial and operational information plays a crucial role in enhancing the transparency of corporate narratives and promoting informed decision-making in emerging markets where information availability is not as extensive.

This study also has implications for market participants. It highlights the importance and impact of disclosure regulations beyond traditional financial information. By focusing on industrial and operational information disclosure regulation, we demonstrate that such regulation plays a crucial role in influencing firms' disclosure practices and enhancing price informativeness. However, the proprietary costs of disclosure can affect the extent to which firms respond to IIDG implementation. Understanding this type of regulation's impact can help investors make informed decisions and better assess the value and risks associated with their investment choices. For regulators and policymakers, our findings emphasize the critical role disclosure regulations play in shaping the information landscape. Thus, designing and implementing effective disclosure regulations that encourage firms to provide relevant and valuable information to investors and other stakeholders can benefit the market.

2. Background, literature review, and hypotheses

2.1. IIDG implementation in China

To standardize the information disclosure in listed firms' annual reports, in 1994 the China Securities Regulatory Commission (CSRC) released its 'Standards on the Content

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and Format of Information Disclosure of Publicly Issued Companies.' These standards, which apply to all firms listed on the Shanghai and Shenzhen stock exchanges, have since been revised several times.

The Shanghai Stock Exchange and the ChiNext board of the Shenzhen Stock Exchange began implementing IIDGs in 2013 to further increase disclosure's usefulness. The Shanghai Stock Exchange targets firms in traditional industries (e.g., real estate, chemicals, and mining), while the ChiNext board of the Shenzhen Stock Exchange focuses on firms in emerging industries (e.g., drugs and biological products, internet games, and video). Unlike the CSRC's general requirements, the IIDGs offer industry-tailored instructions, with the underlying rationale that different industries should provide different types of information. In 2015, the Shenzhen Stock Exchange began establishing a series of IIDGs for firms listed on the exchange.[5] Online Appendix Table I presents detailed information about the specific types of IIDGs and their dates when they became effective. By the end of 2020, the Shanghai Stock Exchange had issued 28 IIDGs, although it abolished 11 of them in January 2021 to streamline and optimize information disclosure practices. During that same period, the Shenzhen Stock Exchange and ChiNext board respectively issued 15 and 18 IIDGs.[6]

The exchanges require listed firms to determine the suitable IIDGs according to their business operations and listed boards and to meet the corresponding information disclosure obligations. The IIDGs specify that applicable firms' annual reports should

[5] In the early 1990s, capital markets were introduced in China with the establishment of the Shanghai Stock Exchange in December 1990 and the Shenzhen Stock Exchange in May 1991. Subsequently, the Shenzhen Stock Exchange established the small and medium enterprise (SME) board in 2004 and the ChiNext board in 2009, which respectively support financing channels for small and middle-sized firms and firms in emerging industries. In 2021, the SME Board were merged into the Main Board of the Shenzhen Stock Exchange.

[6] Of the 28 IIDGs issued by the Shanghai Stock Exchange, 27 are industry-specific IIDGs and one is titled 'General Principles', which specifies the common principles that apply to all firms that must comply with at least one industry-specific IIDG. Given our focus on the information disclosure associated with specific industries, we do not consider the General Principles in the empirical tests.

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include information related to the industry, such as the overall industry development and the firm's potential risk, future development prospects, business models, detailed operational information, and strategic plans and progress. Firms have to disclose the information that the IIDG requires unless they have a legitimate reason for withholding it.[7]

For example, InfoGem Co., Ltd. (Stock code: 300085) is listed on the Shenzhen ChiNext board and should comply with the e-commerce industry's IIDG, which requires that the relevant firms provide information about their business model, platform and website, page views, and registered users. Before the e-commerce IIDG's implementation in 2015, InfoGem did not release this information in detail; after implementation, it substantially increased its MD&A section disclosure. In 2016, the MD&A began to describe its business model as an Original Design Manufacturer (ODM) and provided specific numbers for web page views, visits, and registered users. In Online Appendix Table II, we present keyword lists for four sample IIDGs, along with examples of how firms' disclosure changes around IIDG implementation.

IIDG implementation spurs firms to enhance the quality of their narrative disclosures. By requiring that firms provide industry-related and operational information in their annual reports, IIDGs prompt them to improve the usefulness of their disclosure. In this regard, the MD&A section in annual reports is particularly important. This section contains important narrative information, including IIDG-specified operational details that convey firms' performance, prospects, and strategic objectives to investors and stakeholders.[8] For this reason, when we examine the change in textual disclosure around

[7] The disclosure requirement is intended to enhance the usefulness of the information that listed firms provide. However, exceptional circumstances or legitimate reasons may prevent a firm from fully complying with its disclosure obligations.

[8] Cho and Muslu (2021) state that MD&As represent a rich public depository of corporate narratives, as regulators

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IIDG implementation in the empirical section, we focus on MD&As.

2.2. Literature review

Our research relates to two streams of the literature. The first is research on the economic consequences of disclosure regulations. Numerous studies find that disclosure regulations, such as the entire disclosure regime (e.g., the Sarbanes-Oxley Act (SOX) and mandatory International Financial Reporting Standards (IFRS) adoption), the location of disclosures in financial statements or footnotes, the timing of disclosures, and the restrictions on disclosure discretion, affect market behaviour (Hirst & Hopkins, 1998; Zhang, 2007; Armstrong et al., 2010; deHaan et al., 2015). Increasingly, a growing body of research recognizes the importance of disclosure regulations for non-financial or narrative information, such as CSR or key audit matters (e.g., Chen et al., 2018; Sirois et al., 2018; Burke et al., 2023; Gibbons, 2023). Our study contributes to the literature by investigating how the industry-related information disclosure guidelines implemented in China impact firms' disclosure practices.

The second vein of the literature delves into the determinants of MD&A disclosures and their informativeness. The MD&A section is a critical narrative component of the annual report, and, as such, it allows executives to analyse a firm's historical and prospective operations. It explains fluctuations in key performance indicators, addresses the firm's potential risks and opportunities, and outlines its plans and strategies for the

require that MD&As provide managerial commentary about trends and events that are expected to materially affect the capital market. However, we note that some IIDG-related information may also be disclosed in other portions of firms' annual reports that are narrative in nature, such as financial footnotes. By comparing the changes to MD&As and annual reports, respectively and the similarity between IIDGs and MD&As and annual reports, we find that MD&As are more likely than annual reports to change around IIDG implementation. This finding highlights the importance of the MD&A section as the primary avenue through which firms can comply with industry-specific disclosure requirements. Moreover, when the stock exchanges issue comment letters to firms that do not sufficiently comply with the relevant IIDG, they are more likely to require that the firm provide such information in the MD&A section. See https://www.sohu.com/a/383627722_733484 (in Chinese).

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future. This information is essential to market participants' evaluation of the firm's financial conditions and future cash flows (Barron, 1995; Bryan, 1997; Li, 2010; Sun, 2010; Mayew et al., 2015; Durnev & Mangen, 2020; Cho & Muslu, 2021).

However, as MD&As increase in length and discretion, concerns arise about the usefulness and informativeness of MD&As (Brown & Tucker, 2011). There are three reasons why MD&As may be less informative. First, firms may choose to disclose only template information or information that is vague to avoid criticism or regulatory scrutiny (Jung & Kwon, 1988; Trueman, 1997; Li, 2013; Bertomeu et al., 2021).[9] Second, managerial discretion gives firms opportunities to manipulate the MD&As' content, which can mislead inexperienced investors. For instance, Lo et al. (2017) argue that when firms' reported performance meets or just beats the prior year's performance, they may conceal their poor performance within complex MD&A disclosures. D'Augusta and DeAngelis (2020) present evidence that managers may engage in tone management in the MD&A and that accounting conservatism is negatively related to opportunistic optimism. Third, disclosing proprietary information is associated with certain costs. Such disclosure may draw competitors' attention, potentially eroding a firm's profits and market position. Consequently, managers may hesitate to disclose certain material information to safeguard their competitive advantage (Verrecchia, 1983; Verrecchia & Weber, 2006).

2.3. Hypothesis development

Because the IIDGs encourage firms to provide valuable industry-related and firm-specific information rather than information that merely satisfies market-wide standards, we hypothesize that the policy benefits the market by reducing disclosure processing costs

[9] In theory, firms should disclose all material events. However, many events may be deemed insignificant. Consequently, businesses may choose to not disclose clear information, to disclose only specific types of events, or to manipulate the timing of the disclosure (Li, 2013).

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and limiting managers' discretion in disclosure.

Disclosure processing costs consist of awareness costs, acquisition costs, and integration costs. Awareness costs indicate that it is costly for an investor to learn that the disclosure exists; acquisition costs are those related to obtaining the disclosure and extracting the disclosure; and integration costs involve analyzing the implications of the disclosure for firm value. Such costs can impact investors' information choices, trading behaviours, and overall market outcomes (Blankespoor et al., 2020). Moreover, managerial discretion in disclosure can introduce uncertainty and subjectivity, potentially reducing the disclosed information's usefulness. When firms have the flexibility to selectively disclose information, investors may find it challenging to properly interpret and evaluate the firm's prospects.[10]

By encouraging firms to provide valuable industry-related and firm-specific information, IIDG implementation helps to reduce disclosure processing costs and limit managers' discretion vis-à-vis disclosure by 1) increasing the verifiability of financial and operational information within a firm and 2) making firms in the same IIDG industry more comparable. Both can facilitate investors' integration of information about and correct valuation of the disclosing firms.

First, because IIDG implementation requires firms to provide detailed information about specific aspects of their business (e.g., online sales and shops for retail industry firms or land reverses for the real estate industry), investors can verify the accuracy and consistency of the disclosed financial indicators. For example, if a firm in the retail industry opens new shops over the course of a year, an IIDG requirement that it should

[10] As policymakers state, some firms provide lengthy but uninformative or strategically timed disclosures. Firms can also use too many technical terms or hard-to-understand jargon. Therefore, policymakers' standards should regulate the additional valuable information that listed firms should reveal. See http://www.sse.com.cn/disclosure/credibility/dynamic/c/c_20150918_3990860.shtml (in Chinese).

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disclose the additional operational information (i.e., the openings, locations, areas, and sales) can reveal whether the firm's total sales are consistent with the information about its shops and whether its future profits are expected to further increase. The additional information allows investors to better assess the firm's current state and development plans, thereby lowering their information processing costs and increasing their understanding of the firm's value.

Second, IIDG implementation can help investors understand firm value by making firms in the same IIDG industry more comparable. Using firms in the photovoltaic industry as an example, people can know firms' specific photovoltaic products and their performance or production plans subsequent to IIDG implementation. This increases the availability of firm-specific information, enabling investors to identify the more competitive firms in that industry. Such details were less likely to be available prior to IIDG implementation when investors only had access to aggregated financial data.

The increased information verifiability and comparability induced by IIDGs help investors synthesize and interpret information. Investors can more confidently rely on the disclosed information, which results in reduced information processing costs and better-informed investment decisions. Moreover, IIDG implementation can reduce managerial discretion and promote standardization in firms' disclosure practices, because the IIDG serves as a benchmark with which firms should comply.

When making trading decisions, market participants are more likely to place greater emphasis on IIDG-required disclosure than on other market-wide disclosures if firms provide the information that the relevant IIDG requires, because such information helps investors better understand the firm's value. Consequently, the increased reliance on IIDG-required disclosure in trading decisions leads to an improvement in price informativeness, that is, the extent to which stock prices reflect the available information

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in the disclosure (Brunnermeier, 2005; Blankespoor et al., 2020).

We therefore propose the following hypotheses:

Hypothesis 1: After IIDG implementation, firms that have to follow the IIDGs increase their textual disclosure related to the guidelines more than firms that do not.

Hypothesis 2: After IIDG implementation, price informativeness increases for firms that have to follow the IIDGs relative to those that do not.

However, firms may not provide the information required by the relevant IIDG. If a firm has sufficient reason for being unable or unwilling to disclose IIDG-required information, it disputes the principle of disclosure and must provide a reasonable explanation for why it does not disclose. By providing the explanation, the firm receives an exemption from the disclosure requirement. For example, firms with high proprietary costs may be reluctant to fully comply with the relevant guidelines and therefore maintain more cautious disclosure practices. Thus, we are unlikely to observe a significant change in such firms' textual disclosure and price informativeness after IIDG implementation. We also acknowledge that, in the wake of implementation, not all firms will disclose the same amount of additional information and that the benefits to the market may vary with the amount of additional information that is disclosed.

3. Research design

3.1. Data source and sample selection

Our sample includes all Chinese A-share firms listed on the Shanghai and Shenzhen Stock Exchanges between 2009 and 2019. This sample period covers the three years before the first IIDG adoption year and ends before the 2020 implementation of the new securities

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law.[11] We obtain firms' financial information from the China Stock Market and Accounting Research database (CSMAR). The MD&A texts are from the Chinese Research Data Services database (CNRDS).

We manually identify the relevant IIDG for each firm. Specifically, we first download all the annual reports from CNINFO (http://www.cninfo.com.cn/new/index) and manually search for keywords such as 'industry business analysis,' 'business description,' 'management discussion & analysis,' and 'industry information disclosure guideline' in each annual report. A firm's IIDG is mentioned in the context of these keywords. If a specific IIDG is not explicitly mentioned within this context, we manually read the business description paragraphs in a firm's MD&A to identify the relevant IIDG.

We exclude firm-year observations based on the following criteria: (1) firms in the financial industry; (2) firms that delisted during the sample period; and (3) firm-year observations with missing values. The final sample includes 27,518 firm-year observations with four cohorts for the three years before and three years after each IIDG adoption event (a [-3, +2] year window). Section 3.2 provides more details about how we construct the cohorts. To mitigate the influence of outliers, all continuous variables are winsorized at the 1% and 99% levels.

3.2. Model specification

The IIDGs are implemented in a staggered way. However, criticism of the staggered DiD design is widespread because the presence of treatment effect heterogeneity may result in

[11] The securities law that came into effect in 2020 significantly increases listed firms' disclosure responsibilities and may change how firms disclose information in their 2020 annual reports. In addition, the first IIDGs to be implemented were for the radio, film and television, and pharmaceuticals and biological products industries. These guidelines became effective on January 7, 2013, making 2012 the adoption year. Accordingly, applicable firms' 2012 annual reports should disclose the required information. We also note that we do not use some of the initial IIDGs in the regressions due to stringent data requirements, such as having at least one observation per treated or control firm both before IIDG implementation and after it.

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biased inferences (Goodman-Bacon, 2021; Baker et al., 2022).[12] Baker et al. (2022) show that it is necessary to compare treated firms with a 'clean' set of control firms. Recent studies indicate that the stacked DiD design outperforms the staggered design because the former separately estimates the treatment effects for each implementation event, and the 'unclean' control firms that were already treated earlier are removed in each event study (Gormley & Matsa, 2011; Goodman-Bacon, 2021; Baker et al., 2022). Following Cengiz et al. (2019), we use the stacked DiD design to compare changes in the textual disclosure and price informativeness of the treated and control firms around the IIDG implementation year. In Online Appendix I, we conduct a staggered DiD design and the other methods suggested by Borusyak et al. (2022), Callaway and Sant'Anna (2021), Sun and Abraham (2021), Gardner (2022), and De Chaisemartin and d'Haultfoeuille (2020). The results are consistent with our primary findings.

Specifically, we create four datasets, one for each implementation event cohort, *c*, within which all the treated firms have the same adoption year. For each event, we construct a cohort of the treated firms and all the clean control firms using firm-year observations for the [-3, +2] year window around the event. Clean control firms are those that either remain untreated throughout the sample period (i.e., they are not required to comply with any IIDG regulations) or that are treated at least three years after the relevant adoption event. We require that each treated and control firm have at least one observation both before and after that cohort's IIDG adoption. With this design, some control firms may appear multiple times because they may serve as a control firm for a different cohort, that is, during a different wave of IIDG implementation. Table 1 presents the number of

[12] For example, when the earlier treatment firms serve as the controls for the later treatment firms, the larger change in outcomes for the earlier firms relative to that for the later ones may yield reverse effects.

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observations for the treated and control firms in the four cohorts. Online Appendix Table III shows the specific IIDG for each cohort. We stack these cohorts and estimate the following regression equation so we can examine the relationship between IIDG implementation and textual disclosure that relates to IIDGs:

$$IIDG_related_words_{i,c,t} \;/\; Entityname_ratio_{i,c,t} = \alpha_0 + \alpha_1 Treat_{i,c} * Post_{c,t} + \sum Controls + f_{i,c} + y_{c,t} + \varepsilon_{i,c,t} \quad (1)$$

$IIDG_related_words_{i,c,t}$ is the natural logarithm of 1 plus the number of IIDG-related words in the MD&A of firm i in cohort c in year t. $Entityname_ratio_{i,c,t}$ is the MD&A specificity of firm i in cohort c in year t. $Treat_{i,c}$ is an indicator that is equal to 1 if firm i in cohort c complies with at least one IIDG, and 0 otherwise. $Post_{c,t}$ is an indicator equal to 1 for the adoption year and all the years after it in cohort c, and 0 otherwise. $f_{i,c}$ and $y_{c,t}$ are the vectors for the cohort×firm and cohort×year fixed effects, respectively. The cohort×firm fixed effects control for unobservable, time-invariant firm characteristics for the firms in each cohort, whereas the cohort×year fixed effects control for the time-trending factors related to firms' disclosure behaviour in each cohort. The coefficient on $Treat_{i,c} * Post_{c,t}$ captures the change in textual disclosure (*IIDG_related_words* and *Entityname_ratio*) for the treated firms before and after the implementation events, relative to the control firms in the same cohort and during the same periods. If the IIDG encourages firms to disclose more IIDG-related and firm-specific information in their MD&As, we expect to find that α_1 in Equation (1) will be positive and significant.

Similarly, to examine the relationship between IIDG implementation and price informativeness, we estimate the following regression equation:

$$AbsCAR_{i,c,t} \;/\; Syn_spec_{i,c,t+1} = \alpha_0 + \alpha_1 Treat_{i,c} * Post_{c,t} + \sum Controls + f_{i,c} + y_{c,t} + \varepsilon_{i,c,t} \quad (2)$$

$AbsCAR_{i,c,t}$ is the absolute value of the cumulative abnormal returns over the [-5, +1]

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window, with 0 indicating the release date for the year t annual report of firm i in cohort c. The abnormal return is estimated based on Fama and French's (2015) five-factor model using a prediction window of [-180, -31]. $Syn_spec_{i,c,t+1}$ is the stock return synchronicity of firm i for year t+1 when firm i is in cohort c. If IIDG implementation increases price informativeness, then we expect to find that α_1 in Equation (2) is positive (negative) and significant when the dependent variable is $AbsCAR_{i,c,t}$ ($Syn_spec_{i,c,t+1}$).

3.3. Variable definitions

3.3.1 Textual disclosure in response to IIDGs

We use IIDG-related words in firms' MD&As and the MD&As' specificity to capture the degree to which textual disclosure responds to the relevant IIDG.

(1) IIDG-related words in MD&As (*IIDG_related_words*)

Our first textual measure is the number of IIDG-related words in the MD&As (*IIDG_related_words*). Each IIDG specifies the information that should be disclosed, and different IIDGs have different keywords. For example, firms that comply with the real estate industry IIDG are required to discuss land reserves, construction areas, completed areas, and all their respective details. Meanwhile, firms that comply with the IIDG for the chemical industry are more likely to discuss specific details related to materials, fibers, and production safety. Such specific terms refer to IIDG-related words that are unique to the relevant IIDG. Given that natural language processing packages (i.e., Jieba) may not successfully parse professional and specific terms, we manually extract such terms from each industry guideline. Online Appendix Table II presents the IIDG-related words for the four sample IIDGs. For the treated firms, *IIDG_related_words* is the natural logarithm of 1 plus the number of these words that appear in the MD&A. Because the control firms

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do not need to comply with the cohort's IIDG, we construct the measure as if they had to comply with the same IIDG as the treated firms in the same cohort.[13]

(2) Firm-specific information in MD&As (*Entityname_ratio*)

Our second textual disclosure measure is the specificity of MD&As (*Entityname_ratio*), following Hope et al. (2016). *Entityname_ratio* is equal to the total number of characters in the specific words or phrases that relate to the disclosing firm divided by the total number of characters in the MD&A. We multiply the outcome by 100 to ease the interpretation of the result. Our specific words or phrases consist of five entity categories: (1) the names of organisations (e.g., the China Automotive Industry Association); (2) the names of persons (e.g., Jeme Tien Yow); (3) the names of locations (e.g., Shanxi); (4) the time (e.g., the summer of 2014); and (5) the date (e.g., April 15, 2019). These specific words or phrases are usually firm-specific (Hope et al., 2016). To obtain the entity names, we split the MD&A (excluding tables and figures) into sentences and then identify and extract the specific entity names from each sentence using the Stanford Named Entity Recognition (NER) package. This package provides Chinese models built from the Ontonotes Chinese-named entity data. For example, a sentence in Vanke Co., Ltd.'s (Stock code: 000002) 2017 MD&A states: 'Our firm has signed a strategic cooperation agreement with Banyan Tree Holdings Limited (Chinese name: 悦榕控股有限公司), giving full play to the advantages to both sides in jointly expanding hotels and other related businesses.' Banyan Tree Holdings Limited is identified as the

[13] For the control firms, *IIDG_related_words* is the natural logarithm of 1 plus the number of IIDG-related words that appear in these firms' MD&As, where the IIDG applies to the treated firms in the same cohort. If more than one IIDG applies to a single cohort, we calculate the mean value.

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name of an organisation that seems unlikely to occur in other firms' MD&As.[14] Based on this technique, we can measure how much firm-specific information the MD&A contains.[15]

3.3.2 Price informativeness

We use the market response to the disclosure and stock return synchronicity to proxy for price informativeness.

(1) Absolute cumulative abnormal returns (*AbsCAR*)

We calculate *AbsCAR* around the annual report date using Fama and French's (2015) five-factor model as follows:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + \varepsilon_{i,t} \quad (3)$$

$$AR_{i,t} = (R_{i,t} - R_{f,t}) - (\widetilde{\alpha_i} + \widetilde{\beta_i}(R_{m,t} - R_{f,t}) + \widetilde{s_i} SMB_t + \widetilde{h_i} HML_t + \widetilde{r_i} RMW_t + \widetilde{c_i} CMA_t) \quad (4)$$

$$AbsCAR = |\sum_{t=-5}^{t=1} AR_{i,t}| \quad (5)$$

$R_{i,t}$ is firm i's return on day t, $R_{f,t}$ is the risk-free rate, $R_{m,t}$ is the market return on day t, *SMB* is the difference in returns between small- and large-sized stocks, *HML* is the difference in returns between cheap and expensive stocks, *RMW* is the difference in

[14] Another example is the IIDG for the agricultural, forestry, livestock, and fishery industry. In response to the IIDG, Shandong Homey Aquatic Development Co. Ltd. (stock code: 600467; a firm that has to comply with the IIDG) added in its MD&A: 'To promote research and development, the firm has set up the College of Marine Life Sciences (Chinese name: 海洋生命科学院) and the Marine Food Testing Center (Chinese name: 海洋食品检测中心), and cooperates with the Yellow Sea Fisheries Research Institute of Chinese Academy of Fisheries Sciences (Chinese name: 中国水产科学研究院黄海水产研究所), the Ocean Research Institute of Chinese Academy of Sciences (Chinese name: 中科院海洋研究所), the Shandong Academy of Sciences (Chinese name: 山东省科学院), and the Ocean University of China (Chinese name: 中国海洋大学).' These names can also be identified by the Stanford NER technique as organisation names.

[15] Our analysis does not include the categories of money and percentage that Hope et al. (2016) propose because identifying these two categories is less precise than identifying other categories such as organisation, person, and location (https://nlp.stanford.edu/software/CRF-NER.html). Moreover, we only focus on the MD&A's textual content; money and percentage are more likely to appear in tabular format.

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returns between the most and least profitable stocks, and *CMA* is the difference in returns between low- and high-investment firms. We first use Equation (3) and the prediction window of [-180, -31] trading days relative to the announcement date for an annual report to estimate α_i, β_i, s_i, h_i, r_i, and c_i for each firm. We then use Equation (4) and the predicted $\widetilde{\alpha_i}$, $\widetilde{\beta_i}$, $\widetilde{s_i}$, $\widetilde{h_i}$, $\widetilde{r_i}$, and $\widetilde{c_i}$ to estimate abnormal returns for each day of the event window [-5, +1]. *AbsCAR* is the absolute value of the cumulative abnormal returns over the [-5, +1] window, as shown in Equation (5).

(2) Stock return synchronicity (*Syn_spec*)

Following Morck et al. (2000) and Piotroski and Roulstone (2004), we use the stock return synchronicity to measure how much firm-specific information is incorporated into firms' returns. Specifically, we employ the following models to calculate the synchronicity:

$$RET_{i,t} = \theta_0 + \theta_1 MARET_{t-1} + \theta_2 MARET_t + \theta_3 INDRET_{j,t-1} + \theta_4 INDRET_{j,t} + \varepsilon_{i,t} \quad (6)$$

$$Syn_spec_{i,t} = \text{Ln}\,(R^2 / (1 - R^2)) \quad (7)$$

$RET_{i,t}$ is the buy-and-hold return for firm i in week t. $MARET_t$ is the market return in week t, calculated as the sum of the buy-and-hold returns for each firm weighted by the firm's market value in week t. $INDRET_{j,t}$ is the industry return, calculated as the sum of the buy-and-hold returns for each firm in industry j weighted by the firm's market value in week t. Both calculations exclude firm i. We regress the weekly returns on the current and prior week's value-weighted market and industry returns from May of year t to April of year t+1, as shown in Equation (6). The regression requires at least 30 weekly observations. R^2 from Equation (6) captures how much an individual firm's returns vary with the current and lagged market and industry returns. We then use Equation (7) to

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calculate the log-transformed R^2, denoted as *Syn_spec*. A higher value for *Syn_spec* indicates that more market-wide and industry-wide, but less firm-specific, information is incorporated into a firm's stock returns.

3.3.3 Control variables

Following Piotroski and Roulstone (2004), Gul et al. (2010), Brown and Tucker (2011), Hope et al. (2016), and Kim et al. (2021), we control for a series of firm-level characteristics that may affect MD&A disclosure behaviour and their informativeness. We first control for firm size (*Size*), the market-to-book ratio (*MB*), leverage (*Lev*), the standard deviation of earnings over the previous three years (*Stdroa*), total accruals (|*Accruals*|), and loss (*Loss*). We also control for the MD&A length (*Length_MD&A*) and the length of the annual reports (*Length_AR*). When firms experience a large change in earnings, they may have to offer a more firm-specific explanation for the change or adjust their disclosure strategies. Thus, the model includes the absolute change in earnings per share (|*ΔEPS*|). Additionally, we control for stock return volatility (*Stdret*), the correlation between a firm's profitability and industry performance (*FundCorr*), the change in institutional shareholdings (*ΔInstihold*), institutional shareholdings (*Instihold*), analyst coverage (*Analyst*), the number of firms in the same industry (*Indsize*), the number of subsidiaries (*Subs*), an audit by a Big 4 auditor (*Big4*), the largest shareholder's ownership and its squared value (*Tophold* and $Tophold^2$, respectively). Firms with more volatile returns and more subsidiaries may be required to disclose more details. Additionally, Big 4 auditors, analysts, and institutional shareholders may encourage or pressure a firm to provide more information. All the variables are defined in the appendix.

4. Empirical results

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4.1. Preliminary evidence on firm responses to IIDG implementation

A key assumption underlying our hypotheses is that the treated firms alter their disclosure behaviour in response to IIDG implementation. We validate this assumption in two ways: First, we examine the treated firms' MD&A modification around the implementation. Second, we examine whether the textual similarity between MD&As and the relevant IIDG increases in the adoption years relative to the years before it.

4.1.1 MD&A modification around IIDG implementation

Following Brown and Tucker (2011), we calculate the year-over-year MD&A modification score. Specifically, we use the vector space model (VSM) to form vectors that comprise the words and their frequencies in each MD&A document.[16] We also use the 'total frequency-inverse document frequency' (TF-IDF) to adjust the weight for frequencies of words.[17] We calculate the cosine of the angle between the two vectors for the current- and prior-year MD&A documents as the similarity score for the year-over-year MD&As. The modification score (*MD&A_Modification_Score*) is 1 minus the similarity score, and a higher value indicates a notable difference between consecutive-year MD&A documents.

If the treated firms respond to the policy, we expect a significant increase in MD&A modification in the first adoption year after the implementation. Figure 1(a) shows how the mean and median modification scores for these firms change from the two years before the implementation to the two years after it. The modification score peaks in the

[16] We use the Jieba package to split MD&As. Before splitting an MD&A, we add accounting keywords from the Beijing Sogou Technology Development Co., Ltd., which developed one of the mainstream Chinese input tools (see accounting keywords from https://pinyin.sogou.com/dict/detail/index/20659 (in Chinese)). When splitting each MD&A into words, we remove stop words because they do not have a semantic meaning. We also remove quantitative data (e.g., numbers and percentages) from MD&As to calculate the textual similarity.

[17] By using TF-IDF, the weight of a term (word) increases with the frequency with which it appears in a specific MD&A, but it is penalized when it is common across all MD&As in a year.

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first adoption year (a horizontal axis with zero; *MD&A_Modification_Score* = 0.63) and declines in the years that follow. This provides preliminary evidence that after IIDG implementation, firms revise their MD&As to meet the guideline's requirements.

4.1.2 Textual similarity between firms' MD&As and the relevant IIDGs

If a firm actively complies with the IIDG requirements, we expect the textual similarity between its MD&A and the relevant IIDG to increase when the IIDG becomes effective. To test this prediction, we construct a variable, *Similarity_Score_with_IIDGs*, which is the similarity score between a firm's MD&A and the relevant IIDG. We calculate the score as the cosine of the angle between the two vectors derived from the MD&A and IIDG respectively by applying the same VSM and TF-IDF methods we discuss in the previous section. We standardize the score by subtracting the mean value within a year. Figure 1(b) shows the demeaned scores around IIDG implementation. We find that the textual similarity between a treated firm's MD&A and the relevant IIDG substantially increases in the adoption year, indicating that firms do respond to IIDG implementation.

(Insert Figure 1 here)

4.2. Descriptive statistics and the correlation matrix

We next report the summary statistics for our main tests. Table 1 presents the sample distribution for each of the four cohorts of IIDG implementation. Online Appendix Table III shows the distribution of the treated sample across different IIDG industries. The real estate industry (*SH*) has the highest proportion of firms that apply the IIDG, followed by the chemical industry (SH). Conversely, the internet video (SZ-C) and internet marketing

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(*SZ-C*) industries have the smallest proportion of firms that meet IIDG requirements.[18]

Table 2 presents the descriptive statistics for the variables used in the stacked regressions. The mean for *IIDG_related_words* is 2.576, suggesting that on average, the MD&A includes 12.1 (= $e^{2.576}$-1) IIDG-related words.[19] The mean of *Entityname_ratio* is 4.924, indicating that specific entity names comprise, on average, 4.924 characters per 100 MD&A characters. The average *AbsCAR* is 8.2%. The negative values for the mean and median of *Syn_spec* are consistent with prior literature (e.g., Gul et al., 2010; Jiang et al., 2019; Qiu et al., 2020; Huang et al., 2023) and indicate that the majority of sample firms have an R^2 of less than 50%. The correlation matrix in Online Appendix Table IV shows that the two textual disclosure variables (*IIDG_related_words* and *Entityname_ratio*) are positively correlated. In addition, the absolute values of all the correlation coefficients do not exceed 0.6, indicating that multicollinearity is not a concern in our model.

(Insert Table 1 here)

(Insert Table 2 here)

4.3. The effect of IIDG implementation on textual disclosure

Columns (1)–(4) of Table 3 present the results about the effect of IIDG implementation on textual disclosure. Columns (1) and (3) report the results with only cohort×firm and cohort×year fixed effects included, while Columns (2) and (4) report the results by additionally controlling for firm-level characteristics. The coefficients on *Treat*Post* are

[18] The signifiers *SH*, *SZ*, and *SZ-C* respectively indicate that IIDGs apply to firms listed on the main board of the Shanghai Stock Exchange, the main board of the Shenzhen Stock Exchange, and the ChiNext board of the Shenzhen Stock Exchange.

[19] The mean of IIDG_related_words is low because control firms, which dominate the sample due to the stacked DiD research design, do not need to comply with IIDGs. We thank an anonymous reviewer for pointing this out.

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positive and significant at the 1% level, suggesting that after IIDG implementation, the treated firms not only disclose more IIDG-related information in MD&As but also more firm-specific information. In terms of economic significance, the coefficients on *Treat*Post* in Columns (2) and (4) are 0.349 and 0.432, respectively, suggesting that compared to the average, the number of IIDG-related words for the treated firms increases by 13.5% (= 0.349/2.576) and the percentage of specific entity names for the treated firms increases by 8.8% (= 0.432/4.924).

In terms of the control variables, we find that an MD&A's length is positively associated with words related to IIDGs, while the length of the annual reports is positively correlated with the number of characters in specific entity names scaled by the total number of characters in the MD&A. Firms with more subsidiaries tend to incorporate more IIDG-related words and specific entity names into their MD&As because they are more likely to describe their subsidiaries' business operations in detail (e.g., the operating areas and specialized terms related to their business).

4.4. The effect of IIDG implementation on price informativeness

The results of Equation (2) are reported in Table 3, Columns (5)–(8). The positive and significant coefficients on *Treat*Post* in Columns (5) and (6) suggest that IIDG implementation enhances the market response to the disclosure of firm information. Concerning the economic significance, the result in Column (6) indicates that after implementation, the market reaction increases by 11% (= 0.009/0.082) for the treated firms relative to the control firms. The negative and significant coefficients on *Treat*Post* in Columns (7) and (8) suggest that IIDG implementation increases the incorporation of firm-specific information into stock returns. In terms of economic significance, the result in Column (8) indicates that an average firm's stock return synchronicity decreases from

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-0.259 to -0.354, which means that 43.6% (= $e^{-0.259}/(1+e^{-0.259})$) and 41.2% (= $e^{-0.354}/(1+e^{-0.354})$) of the stock returns can be respectively explained by market-wide and industry-level returns. This indicates that after IIDG implementation, the stock returns that can be explained by firm-specific information increase by 2.4% (= 43.6%–41.2%).[20]

(Insert Table 3 here)

4.5. Identifying causality via the 2SLS method and subsample tests

To further verify that the increase in price informativeness is driven by the change in textual disclosure related to IIDG implementation, we first use the two-stage least squares (2SLS) method. Specifically, we utilize *Treat*Post* as the instrumental variable for the textual disclosure variables (*IIDG_related_words* and *Entityname_ratio*) and then investigate the association between the instrumented textual disclosure variables (*IIDG_related_words* (*instrumented*) and *Entityname_ratio* (*instrumented*)) and the price informativeness variables (*AbsCAR* and *Syn_spec*).[21] The results are shown in Table VI of Online Appendix I. We find that both the instrumented textual disclosure variables are significantly and positively (negatively) associated with the market reaction (stock return synchronicity), suggesting that the IIDG-induced change in textual disclosure contributes to the change in price informativeness.

Second, we divide our sample into two based on the degree of change in textual

[20] The result does not change when we calculate *AbsCAR* using alternative event windows. Online Appendix Table V offers further evidence of the information usefulness associated with IIDG implementation by examining its effect on the verifiability of a firm's financial and operational information and its comparability with firms in the same IIDG industry. Verifiability is captured by the firm's financial quality, which is the signed discretionary accruals based on the modified Jones model. We determine comparability by the mean textual similarity of a firm's MD&As to those of other firms that must comply with the same IIDG. We find that after IIDG implementation, both the verifiability and comparability of these firms' information increases relative to those of the control firms.

[21] The two textual disclosure variables may be endogenous and exposed to IIDG implementation events. As Hudson et al. (2017) suggest, the instrument variable estimate is a properly weighted average of the causal effects of textual disclosure in response to IIDGs for firms whose textual disclosure status is affected by the instrument.

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disclosure. We find that IIDG implementation will result in higher price informativeness if the textual disclosure changes more in response to the IIDG. The result is presented in Table VII of Online Appendix I.

4.6. The parallel trend and other robustness tests

We present the parallel trend test by replacing the key variable of interest, *Treat*Post*, with variables that indicate the treated firms and the two years prior to IIDG adoption (*Treat**(*Year* = *t*-2)), one year before the adoption (*Treat**(*Year* = *t*-1)), the adoption year (*Treat**(*Year* = *t*)), one year after adoption (*Treat**(*Year* = *t*+1)), and two years post adoption (*Treat**(*Year* = *t*+2)).[22] Table VIII of Online Appendix I presents the dynamic effects of IIDG implementation. Online Appendix Figure I presents the estimated coefficients that show that the parallel trend assumption is not violated.

To alleviate endogeneity concerns, we report a series of robustness tests, including (1) a placebo test; (2) a test that exploits the staggered DiD design and alternative methods for alleviating the concern about the staggered DiD design; (3) a test that uses the PSM method; and (4) tests that use alternative measures. We continue to find that our main results are robust to these tests. Online Appendix I presents the details.

5. Additional tests

5.1. Consequences for firms with fewer changes to their textual disclosure

Thus far, our findings indicate that firms offer more information and that the stock market benefits from IIDG implementation. Given that the IIDGs do not stipulate substantial

[22] In the new models, the textual disclosure and price informativeness in the three years before adoption serve as the benchmark; the sample period is the [-3, +2] year window.

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penalties for non-compliance, a natural question is why firms would choose to comply. Despite the absence of prescribed penalties, we contend that non-compliant firms nevertheless experience regulatory consequences.

Specifically, we examine whether firms with fewer changes to their textual disclosure after IIDG implementation are more prone to attracting regulatory scrutiny, as evidenced by disclosure-related comment letters. Because China began to publicly disclose comment letters about 2014 annual reports in 2015, we restrict our sample to the period 2014–2019. We also require that the comment letters mention the firms' disclosure practices. We construct a dummy variable, *Inquiry*, which equals 1 if a firm receives a disclosure-related comment letter about its annual report, and 0 otherwise. We then construct a variable, *Change in Inquiry*, which is equal to the change in the average for *Inquiry* for the three years including and after the adoption year relative to the three years before the adoption year.

To reflect the change in textual disclosure, we construct a dummy variable, *Fewer increases in IIDG_related_words* (*Fewer increases in Entityname_ratio*) is equal to 1 if the change in the average for *IIDG_related_words* (*Entityname_ratio*) for the three years including and after the adoption year relative to the three years before the adoption year is in the bottom quintile, and 0 otherwise. We also construct a dummy variable, *Fewer increases in IIDG_related_words & Fewer increases in Entityname_ratio* that is equal to 1 if both *Fewer increases in IIDG_related_words* and *Fewer increases in Entityname_ratio* are equal to 1, and 0 otherwise.

By focusing on the adoption year sample, we examine whether firms with fewer changes in their textual disclosure (*Fewer increases in IIDG_related_words & Fewer increases in Entityname_ratio*, *Fewer increases in IIDG_related_words*, and *Fewer increases in Entityname_ratio*) are more likely to receive disclosure-related comment

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letters (*Change in Inquiry*). We control for the changes in the control variables and include cohort fixed effects. The results are shown in Table 4. The coefficients on *Fewer increases in IIDG_related_words & Fewer increases in Entityname_ratio*, *Fewer increases in IIDG_related_words*, and *Fewer increases in Entityname_ratio* are all positive; two of them are also significant. These outcomes suggest that the exchanges may pay more attention to firms that make fewer changes to their textual disclosure in response to IIDGs.

(Insert Table 4 here)

5.2. Cross-sectional tests

In this section, we examine whether there is cross-sectional variation in the effects of IIDG implementation on firms' textual disclosure and price informativeness.

Given that numerous previous studies suggest that firms' disclosure is highly related to either industry concentration and competition (Verrecchia, 1983; Ali et al., 2014; Huang et al., 2017), we first examine whether industry concentration or the competitive environment affects firms' incentives to disclose under conditions of regulatory advocacy. Due to the proprietary costs of disclosure, firms in more concentrated industries are less likely to disclose firm-specific information, as doing so could attract new entrants and erode industry incumbents' profits. Therefore, we expect the textual disclosure in response to IIDGs to be less pronounced for these firms.

Following Verrecchia and Weber (2006), we use the Herfindahl index (*HHI*) to proxy for the industry-level concentration. *HHI* is equal to 1 if the Herfindahl index for a firm's industry is above the median for the cohort and year, and 0 otherwise. Firms with an *HHI* equal to 1 operate in a more concentrated industry. These concentrated industries may possess valuable proprietary information that reinforces barriers to entry. We then

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incorporate the industry concentration measure together with its interactions with *Treat*, *Post*, and *Treat*Post* into Equation (1). Columns (1) and (2) of Table 5, Panel A present our findings. The significant and negative coefficients on *Treat*Post*HHI* indicate that the effect of IIDG implementation on textual disclosure is less pronounced for firms in more concentrated industries.[23]

We then investigate whether the firm-level proprietary costs of disclosure, as opposed to the industry-level costs, could also weaken the positive relationship between IIDG implementation and textual disclosure. Specifically, we use the profit margin (*Margin*) to proxy for a firm's proprietary costs. *Margin* is equal to 1 if a firm's operating profit scaled by its total sales is above the median for the cohort and year, and 0 otherwise. The results are shown in Panel A of Table 5, Columns (3) and (4). We find that after IIDG implementation, firms with higher profit margins are less likely to include IIDG-related words in their disclosure relative to those with lower profit margins, suggesting that higher proprietary costs induce these firms to avoid disclosing more detailed information.

Lastly, Panel B of Table 5 examines whether these factors play a role in the effect of IIDG implementation on price informativeness. We do not find significant results, possibly because the stock market has yet to fully recognize the potential impediments associated with IIDG implementation.

(Insert Table 5 here)

6. Conclusion

Information disclosure is an important topic for both practitioners and researchers. It is crucial to understand how to enhance the value of information disclosure, which would

[23] The coefficients on *X* and *Treat*X* (X denotes *HHI*) are not shown in the regression in Columns (1) and (2). Since *X* is an industry-level indicator, both *X* and its interaction with *Treat* are absorbed by cohort×firm fixed effects.

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thereby improve the quality of the capital markets. Staggered IIDG implementation in China offers an ideal setting for a natural experiment that investigates the impact of the regulation of industry information disclosure on price informativeness, specifically through the channel of textual disclosure.

Our findings indicate that after IIDG implementation, firms include more IIDG-related words and more firm-specific information in their MD&As. Additionally, IIDG implementation increases the market reaction to the release of firm information and facilitates the incorporation of firm-specific information into individual firms' stock returns. Our results are robust to the parallel trend test, the placebo test, and the PSM method. Further analysis reveals that applicable firms are less likely to disclose IIDG-related and firm-specific information when they are more likely to attract new entrants or have higher profit margins. Moreover, the probability of receiving a comment letter about disclosure issues increases if a firm's MD&As add less IIDG-related and firm-specific information after IIDG implementation.

Overall, our findings suggest that requiring firms to disclose information according to IIDGs contributes to the interpretation of firm value and brings additional benefits to the market. Our findings should be of interest to market information users (such as employees, investors, regulators, analysts, and auditors). Future research could discuss other economic consequences of information disclosure regulation, such as audit behaviour or spillover effects.

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Supplemental Data and Research Materials

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Appendix

Variable definitions

Variable	Definition
Dependent variables:	
$IIDG_related_words_{i,c,t}$	The natural logarithm of 1 plus the number of IIDG-related terms mentioned in the MD&A for treated firm *i* in cohort *c* in year *t*, when the IIDG applies to firm *i*. For control firm *i* in cohort *c* in year *t*, $IIDG_related_words_{i,c,t}$ is the natural logarithm of 1 plus the number of IIDG-related terms mentioned in the firm's MD&A, where the IIDG applies to the treated firms in the same cohort. If more than one IIDG applies to a single cohort, we calculate the mean value of the natural logarithms.
$Entityname_ratio_{i,c,t}$	The number of characters in the entity names in the MD&A for firm *i* in cohort *c* in year *t* scaled by the total number of characters in the MD&A and multiplied by 100. Entity names are identified based on the Stanford NER technique and comprise five categories: the names of organizations, persons, locations, times, and dates.
$AbsCAR_{i,c,t}$	Absolute value of the cumulative abnormal returns over the [-5, +1] window around the release date of the annual reports for firm *i* in cohort *c* in year *t*.
$Syn_spec_{i,c,t}$	Stock return synchronicity for firm *i* in cohort *c* in year *t*, calculated as the natural logarithm of $(R^2/(1 - R^2))$, where R^2 is estimated by regressing firm *i*'s weekly returns on the current and lagged market and industry weekly returns during the period from May of year *t* to April of year *t*+1.
Key independent variables:	
$Treat_{i,c}$	Indicator equal to 1 if firm *i* in cohort *c* has to comply with one IIDG during the sample period, and 0 otherwise.
$Post_{c,t}$	Indicator equal to 1 if the annual report for year *t* of the treated firms in cohort *c* must be disclosed according to the IIDG, and 0 otherwise.
Control variables:	
$Size_{i,c,t}$	Natural logarithm of the total assets for firm *i* in cohort *c* in year *t*.
$MB_{i,c,t}$	The market value of equity divided by the book equity for firm *i* in cohort *c* in year *t*.
$Lev_{i,c,t}$	The total liabilities divided by the total assets for firm *i* in cohort *c* in year *t*.
$Stdroa_{i,c,t}$	The standard deviation of ROA over the last three years

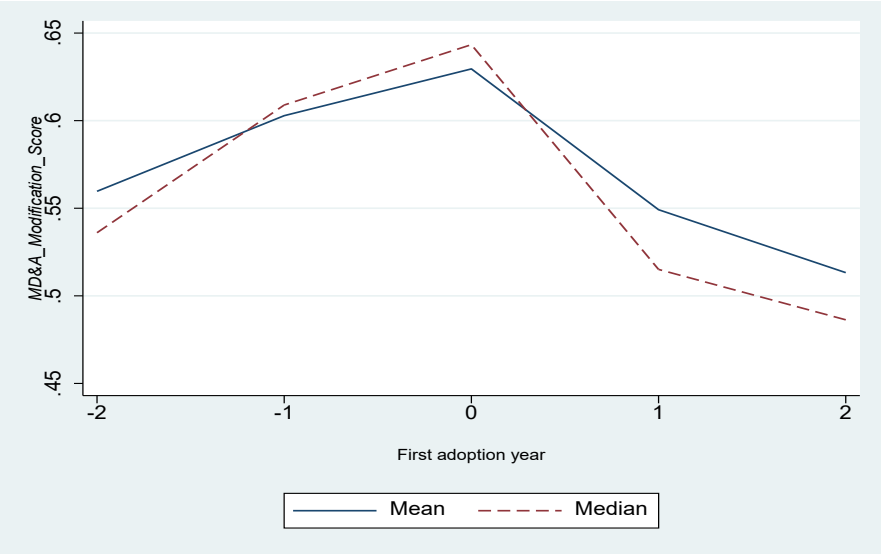
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	for firm *i* in cohort *c* in year *t*, where ROA is the operating income divided by total assets.
$\lvert \Delta EPS_{i,c,t} \rvert$	Absolute value of the difference between the net income per share in year *t* and that in year *t*-1 for firm *i* in cohort *c*.
$\lvert Accruals_{i,c,t} \rvert$	Absolute value of accruals, calculated as operating income minus net operating cash flows for firm *i* in cohort *c* in year *t*, scaled by the lagged total assets.
$Loss_{i,c,t}$	Indicator equals 1 if the net income minus the non-recurrent gains and losses is less than 0 for firm *i* in cohort *c* in year *t*, and 0 otherwise.
$Length_MD\&A_{i,c,t}$	Natural logarithm of the total number of characters in the words in the MD&A for firm *i* in cohort *c* in year *t*.
$Length_AR_{i,c,t}$	Natural logarithm of the total number of characters in the words in the annual report for firm *i* in cohort *c* in year *t*.
$Stdret_{i,c,t}$	The standard deviation of the monthly stock returns over the previous three years for firm *i* in cohort *c* in year *t*.
$FundCorr_{i,c,t}$	Correlation of earnings, calculated as the log-transformed R^2 from regressing firms' ROA on the industry value-weighted ROA in the prior 12 quarters for firm *i* in cohort *c* in year *t*.
$\Delta Instihold_{i,c,t}$	Change in institutional shareholders' share percentage in firm *i* in cohort *c* in year *t*.
$Instihold_{i,c,t}$	Institutional shareholders' share percentage in firm *i* in cohort *c* in year *t*.
$Analyst_{i,c,t}$	Natural logarithm of 1 plus the number of analysts covering firm *i* in cohort *c* in year *t*.
$Indsize_{j,c,t}$	Natural logarithm of the number of firms in industry *j* that are in cohort *c* in year *t*.
$Subs_{i,c,t}$	Natural logarithm of 1 plus the number of subsidiaries for firm *i* in cohort *c* in year *t*.
$Big4_{i,c,t}$	Indicator that equals 1 if firm *i* in cohort *c* in year *t* is audited by a Big 4 auditor, and 0 otherwise.
$Tophold_{i,c,t}$	The largest shareholder's share percentage in firm *i* in cohort *c* in year *t*.
$Tophold_{i,c,t}^2$	Square of the largest shareholder's share percentage in firm *i* in cohort *c* in year *t*.

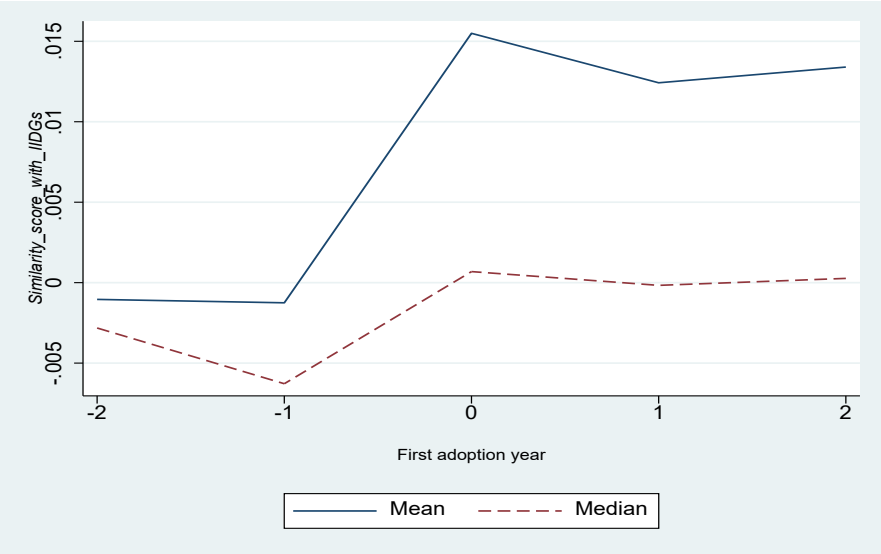
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Figure 1. Preliminary evidence on firm responses to IIDG implementation

(a) Firms' year-over-year MD&A modification around IIDG implementation



(b) Textual similarity between firms' MD&As and the relevant IIDG around the IIDG's implementation



Note: Figure 1(a) shows firms' year-over-year MD&A modification around IIDG implementation, with the vertical axis indicating the modification score (*MD&A_Modification_Score*), which we calculate based on consecutive-year MD&A documents (the current and prior years). Figure 1(b) shows the textual similarity between firms' MD&As and the relevant IIDG around IIDG implementation, with the vertical axis indicating the demeaned textual similarity score (*Similarity_Score_with_IIDGs*). In both figures, a zero on the horizontal axis indicates the first year that the firm's annual report must be disclosed according to the relevant IIDG. The solid and dotted lines are the mean and median scores, respectively.

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Table 1. Distribution of the firms in each cohort for the stacked sample

	Cohort 1		Cohort 2		Cohort 3		Cohort 4	
	Adoption year=2013		Adoption year=2015		Adoption year=2016		Adoption year=2017	
	Treat=0	*Treat*=1	*Treat*=0	*Treat*=1	*Treat*=0	*Treat*=1	*Treat*=0	*Treat*=1
Post=1	2,778	347	2,806	995	2,841	408	2,959	162
Post=0	2,952	326	2,886	1,017	3,115	419	3,343	164
Total	**5,730**	**673**	**5,692**	**2,012**	**5,956**	**827**	**6,302**	**326**

Note: This table presents the distribution of the firms in each cohort for the stacked sample.

Table 2. Descriptive statistics

Variable	N	Mean	Median	S.D.	min	Q1	Q3	max
IIDG_related_words	27,518	2.576	2.448	1.037	0.536	1.782	3.312	5.252
Entityname_ratio	27,518	4.924	4.532	2.768	0.000	3.045	6.393	14.406
AbsCAR	27,518	0.082	0.056	0.086	0.001	0.025	0.107	0.491
Syn_spec	27,518	-0.259	-0.193	0.861	-2.572	-0.821	0.370	1.474
Treat	27,518	0.139	0.000	0.346	0.000	0.000	0.000	1.000
Post	27,518	0.517	1.000	0.500	0.000	0.000	1.000	1.000
Size	27,518	22.251	22.146	1.202	19.517	21.442	22.953	25.649
MB	27,518	3.074	2.109	3.295	0.298	1.322	3.473	21.374
Lev	27,518	0.453	0.447	0.204	0.066	0.297	0.606	0.943
Stdroa	27,518	0.028	0.017	0.035	0.001	0.008	0.033	0.202
|ΔEPS|	27,518	0.246	0.117	0.363	0.001	0.042	0.284	2.258
|Accruals|	27,518	0.062	0.044	0.065	0.001	0.020	0.080	0.352
Loss	27,518	0.218	0.000	0.413	0.000	0.000	0.000	1.000
Length_MD&A	27,518	8.724	8.729	0.444	7.555	8.433	9.019	9.784
Length_AR	27,518	11.864	11.860	0.246	11.264	11.707	12.018	12.517
Stdret	27,518	0.029	0.027	0.010	0.013	0.022	0.033	0.057
FundCorr	27,518	-2.676	-2.112	2.007	-10.021	-3.560	-1.208	-0.257
ΔInstihold	27,518	0.000	0.000	0.053	-0.155	-0.023	0.020	0.172
Instihold	27,518	0.063	0.041	0.067	0.000	0.012	0.094	0.314
Analyst	27,518	1.462	1.386	1.155	0.000	0.000	2.398	3.689
Indsize	27,518	5.813	6.172	1.188	2.485	4.920	6.820	7.251
Subs	27,518	2.524	2.565	0.968	0.000	1.946	3.135	4.942
Big4	27,518	0.057	0.000	0.233	0.000	0.000	0.000	1.000
Tophold	27,518	0.343	0.323	0.147	0.088	0.226	0.441	0.743
Tophold^2	27,518	0.139	0.104	0.115	0.008	0.051	0.194	0.552

Note: This table presents the summary statistics for the variables used in the main regressions. All the variables are defined in the appendix, and all continuous variables are winsorized at the 1st and 99th percentiles.

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Table 3. The effect of IIDG implementation on textual disclosure and price informativeness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IIDG_related_words	*IIDG_related_words*	*Entityname_ratio*	*Entityname_ratio*	*AbsCAR*	*AbsCAR*	*Syn_spec*	*Syn_spec*
*Treat*Post*	**0.496*** **	**0.349*** **	**0.439*** **	**0.432*** **	**0.008*** **	**0.009*** **	**-0.100*** **	**-0.095*** **
	(18.05)	**(15.49)**	**(4.57)**	**(4.29)**	**(2.76)**	**(2.93)**	**(-3.45)**	**(-3.21)**
Size		0.028**		-0.081		-0.012***		0.176***
		(2.16)		(-0.84)		(-3.51)		(7.10)
MB		-0.001		-0.011		0.000		0.022***
		(-0.54)		(-0.77)		(0.35)		(5.28)
Lev		-0.004		0.525*		0.033***		-0.387***
		(-0.11)		(1.72)		(2.65)		(-4.68)
Stdroa		0.140		1.909		0.034		-0.223
		(1.24)		(1.64)		(0.93)		(-0.72)
|ΔEPS|		-0.018*		0.144		0.005		0.010
		(-1.78)		(1.57)		(1.55)		(0.41)
|Accruals|		-0.014		-0.489		0.031*		-0.144
		(-0.32)		(-1.16)		(1.79)		(-1.17)
Loss		-0.028***		-0.103		0.004		-0.092***
		(-3.38)		(-1.41)		(1.23)		(-3.83)
Length_MD&A		0.773***		0.008		0.000		-0.020
		(55.35)		(0.08)		(0.11)		(-0.71)
Length_AR		0.025		1.187***		-0.028***		0.127*
		(0.85)		(4.27)		(-2.93)		(1.76)
Stdret		0.351		2.658		-0.232		9.915***
		(0.69)		(0.56)		(-1.03)		(6.40)
FundCorr		-0.000		-0.017		0.000		-0.004
		(-0.07)		(-1.37)		(0.98)		(-1.12)

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ΔInstihold		-0.068		0.158		-0.000		-0.061
		(-1.33)		(0.32)		(-0.02)		(-0.40)
Instihold		0.059		0.884		0.073***		-0.837***
		(0.80)		(1.31)		(2.79)		(-4.23)
Analyst		-0.011***		0.044		-0.002		0.048***
		(-2.73)		(1.19)		(-1.55)		(3.98)
Indsize		-0.004		0.382		0.007		-0.004
		(-0.07)		(0.84)		(0.52)		(-0.03)
Subs		0.033***		0.217***		0.002		-0.014
		(3.63)		(3.12)		(0.79)		(-0.84)
*Big*4		-0.059*		-0.014		-0.004		0.090
		(-1.78)		(-0.06)		(-0.42)		(0.98)
Tophold		0.308*		-0.843		0.042		0.275
		(1.65)		(-0.53)		(0.69)		(0.56)
$Tophold^2$		-0.071		0.574		-0.016		-1.107*
		(-0.29)		(0.30)		(-0.22)		(-1.80)
Constant	2.541***	-5.245***	4.893***	-10.498***	0.081***	0.616***	-0.252***	-5.559***
	(1,322.13)	(-10.46)	(727.75)	(-2.60)	(400.42)	(4.27)	(-124.47)	(-5.09)
Cohort*Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Cohort*Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	27,518	27,518	27,518	27,518	27,518	27,518	27,518	27,518
R-squared	0.914	0.944	0.593	0.597	0.266	0.273	0.506	0.518
F	325.7	191.5	20.88	4.688	7.634	4.035	11.92	12.18

Note: This table presents the results about the effect of IIDG implementation on textual disclosure and price informativeness. *IIDG_related_words* is the natural logarithm of 1 plus the number of IIDG-related terms that appear in the MD&A for the treated firms. For the control firms, *IIDG_related_words* is the natural logarithm of 1 plus the number of IIDG-related terms mentioned in these firms' MD&As for the IIDG that is relevant to the treated firms in the same cohort. If more than one IIDG applies to a single cohort, we calculate the mean value. *Entityname_ratio* is the number of characters in the entity names in the MD&A scaled by the total number of characters in the MD&A and multiplied by 100. *AbsCAR* is the absolute value of the cumulative abnormal returns over the [-5, +1] window around the release date of the annual reports. *Syn_spec* is a measure of the stock return synchronicity, calculated as the natural logarithm of $(R^2/(1 - R^2))$, where R^2 is estimated by regressing a firm's weekly returns on the current and lagged market and industry weekly returns during the period from May of the current year to April of the following year. All other variables are defined in the appendix. We report t-statistics based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

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Table 4. Consequences for firms with fewer changes to their textual disclosure

	(1)	(2)	(3)
	Change in Inquiry	*Change in Inquiry*	*Change in Inquiry*
Fewer increases in IIDG_related_words & Fewer increases in Entityname_ratio	**0.048****		
	(2.23)		
Fewer increases in IIDG_related_words		0.006	
		(0.59)	
Fewer increases in Entityname_ratio			**0.019***
			(1.94)
Controls	YES	YES	YES
Cohort FE	YES	YES	YES
Observations	3,324	3,324	3,324
R-squared	0.081	0.078	0.080
F	4.486	4.321	4.573

Note: This table presents the results about the consequences for firms that make fewer changes to their textual disclosure. *Change in Inquiry* is equal to the change in the average for *Inquiry* for the three years including and after the adoption year relative to the three years before the adoption year, and 0 otherwise. *Inquiry* is equal to 1 if a firm receives a disclosure-related comment letter on the annual report, and 0 otherwise. *Fewer increases in IIDG_related_words* (*Fewer increases in Entityname_ratio*) is equal to 1 if the change in the average for *IIDG_related_words* (*Entityname_ratio*) for the three years including and after the adoption year relative to the three years before the adoption year is in the bottom quintile of the cohort, and 0 otherwise. *Fewer increases in IIDG_related_words* & *Fewer increases in Entityname_ratio* is equal to 1 if both *Fewer increases in IIDG_related_words* and *Fewer increases in Entityname_ratio* are equal to 1, and 0 otherwise. The control variables are also the change values. We report t-statistics based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

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Table 5. Cross-sectional tests for the effects of IIDG implementation

Panel A: Cross-sectional tests for the effects of IIDG implementation on textual disclosure				
	(1)	(2)	(3)	(4)
	X=*HHI*		X=*Margin*	
	IIDG_related_words	*Entityname_ratio*	*IIDG_related_words*	*Entityname_ratio*
*Treat*Post*	0.499***	1.292***	0.527***	0.500*
	(6.57)	(4.25)	(7.74)	(1.74)
X			-0.010	0.016
			(-0.93)	(0.15)
*Treat*X*			0.172***	0.047
			(3.57)	(0.25)
*Post*X*	0.008	-0.013	-0.006	-0.001
	(0.80)	(-0.15)	(-0.77)	(-0.01)
*Treat*Post*X*	**-0.092****	**-0.525*****	**-0.120*****	**-0.037**
	(-2.01)	**(-2.80)**	**(-2.88)**	**(-0.21)**
Controls	YES	YES	YES	YES
Cohort*Firm FE	YES	YES	YES	YES
Cohort*Year FE	YES	YES	YES	YES
Observations	27,518	27,518	27,510	27,510
R-squared	0.944	0.598	0.945	0.599
F	178.4	4.936	162.6	4.216
Panel B: Cross-sectional tests for the effects of IIDG implementation on price informativeness				
	(1)	(2)	(3)	(4)
	X=HHI		*X=Margin*	
	AbsCAR	*Syn_spec*	*AbsCAR*	*Syn_spec*
*Treat*Post*	0.013	-0.042	0.009	0.069
	(1.33)	(-0.44)	(0.97)	(0.75)
X			-0.001	0.011
			(-0.14)	(0.37)
*Treat*X*			0.004	0.059
			(0.53)	(1.04)
*Post*X*	0.002	0.019	-0.001	0.035
	(0.56)	(0.80)	(-0.40)	(1.59)
*Treat*Post*X*	**-0.003**	**-0.035**	**-0.000**	**-0.109***
	(-0.46)	**(-0.60)**	**(-0.00)**	**(-1.85)**
Controls	YES	YES	YES	YES
Cohort*Firm FE	YES	YES	YES	YES
Cohort*Year FE	YES	YES	YES	YES
Observations	27,518	27,518	27,510	27,510
R-squared	0.273	0.518	0.273	0.518
F	3.710	11.08	3.305	10.24

Note: This table presents the cross-sectional tests for the effects of IIDG implementation on textual disclosure and price informativeness. *HHI* is equal to 1 if the Herfindahl index of a firm's industry is above the median for the cohort and year, and 0 otherwise. *Margin* is equal to 1 if a firm's operating profit scaled by its total sales is above the median for the cohort and year, and 0 otherwise. All other variables are defined in the appendix. We report t-statistics based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

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